

1 Introduction

This document will contain the implementation details for the design learning platform. It will be used to outline the implementation before it is developed.

2 Components

In this section, I will outline the relationship between the following components and the practical steps needed to establish the relationship:

1. Artha - Meaningful Programming Language
2. DAL - Design Abstraction Language
3. CSM - Computable Semantic Model
4. CSP - Computable Semantic Primitives
5. Engine - Semantic Reasoning Engine
6. SI - Synthesized Implementation

When discussing this framework, the core principles revolve around meaning. So I think I will first start by first establishing that meaning is first constructed by narratives that use the primitive meaningful building blocks. The narratives themselves have a structure and that structure is used to build a semantic model written in the design abstraction language. Since the meaning was built from computable semantic primitives, which are themselves established using the DAL, the semantic model becomes executable and becomes a computable semantic model.

Once meaning has been established, now it can be used in another narrative. It is a compositional system where meaning is composed through narratives and used to compose more meaning. Since the meaning is built from computable semantic primitives and it has an unambiguous structure, it can be synthesized into an implementation. In the process, the meaning of the software system has been automatically developed in a programming language.

The engine reasons about the meaning using the structure established by the computable semantic model. Since the structure is unambiguous and meaningful, it is able to reason about it to enforce its own definition of correctness and to also identify missing or incorrect meaning. Through the synthesis and instrumentation, the necessary information needed to replay the execution is preserved and the engine is able to diagnose any failures and enable the meaning to expand through root cause analysis. This process automatically manages the software system and enables a deterministic learning loop.

Next, I will discuss how each component can be practically implemented.